



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION II

CANDIDATE
NAME

CENTRE
NUMBER

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CLASS

INDEX
NUMBER

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H2 BIOLOGY

9744/04

Paper 4 Practical

28 Aug 2017

Candidates answer on the Question Paper.

2 hours 30 minutes

Additional Materials: As listed in the Apparatus and Materials List.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your class, index number and name on all the work you hand in.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You will lose marks if you do not show your working, or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

Shift	
Laboratory	
For Examiner's Use	
1	/ 22
2	/ 19
3	/ 14
Total	/ 55

Answer **all** questions.

Question 1

In this question, you will investigate the effect of glucose and sucrose on the rate of respiration in yeast cells.

You are provided with:

- 40 cm³ yeast cell suspension, labelled **Y**
- 25 cm³ blue alkaline solution, labelled **C**
- 25 cm³ hydrochloric acid, labelled **H**
- 20 cm³ of 0.2 mol dm⁻³ glucose solution, labelled **G**
- 20 cm³ of 0.2 mol dm⁻³ sucrose solution, labelled **S**
- 20 cm³ of distilled water, labelled **W**

Read through steps 1 to 12 and prepare a table to record your results in step (c), before starting the investigation.

Stage 1

Perform the following steps to activate the yeast cells in **Y** to prepare for **stage 3**

- 1** Label 3 boiling tubes **B1**, **B2**, and **B3**. Add 10 cm³ of **Y** into each of the boiling tubes.
- 2** Add 10 cm³ of **G**, **S**, and **W** into boiling tubes **B1**, **B2**, and **B3**, respectively.
- 3** Start the stopwatch. Leave the tubes aside for at least 15 minutes, before beginning on **stage 3**. During this time, you should attempt the rest of the question.

Stage 2

C, an alkaline solution, is blue because it contains an indicator.

Carbon dioxide reacts with **C** when bubbled into it. When enough carbon dioxide reacts with **C**, the indicator will change from blue to yellow (even if the mixture is cloudy). This is the end-point.

If only a small volume of carbon dioxide is bubbled into **C** then the indicator will remain blue. The end-point can then be reached by adding hydrochloric acid, **H**, slowly until the indicator turns from blue to yellow. The volume of **H** is then recorded.

The volume of **H** added to reach the end-point indicates how much carbon dioxide had reacted with **C**. The lesser the amount of carbon dioxide is bubbled into **C**, the more volume of **H** will need to be added to **C** to get the end-point.

You are required to find the volume of **H** needed to reach the end-point when **no** carbon dioxide has been bubbled into **C**.

- 4** Put 5 cm³ of **C** into a test-tube.

- 5** Use a 6 cm³ syringe, containing 3 cm³ of **H**, to put drops of **H** into **C** as shown in

Fig. 1.1. Mix well as you add **H**, until the end-point is reached. You may need to fill the syringe again.

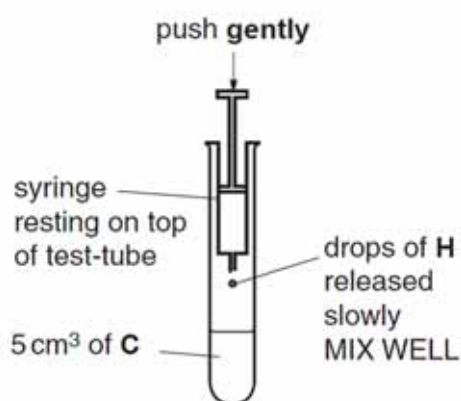


Fig. 1.1

(a) Record the volume of **H** needed to reach the end-point.

[1]

PDO

1. **Correct value;**
Acceptable range 1.0 – 2.0 cm³
2. **Precision;**
1 d.p.

Volume of **H** needed = _____ cm³

Stage 3

You are required to investigate the effect of glucose and sucrose on the release of carbon dioxide from a yeast cell suspension using apparatus set up as in Fig.1.2, and the tubes prepared in **stage 1**.

Yeast cells release carbon dioxide from some of their metabolic reactions when provided with respiratory substrates. Tube **B1** contains glucose as the respiratory substrate, whilst tube **B2** contains sucrose.

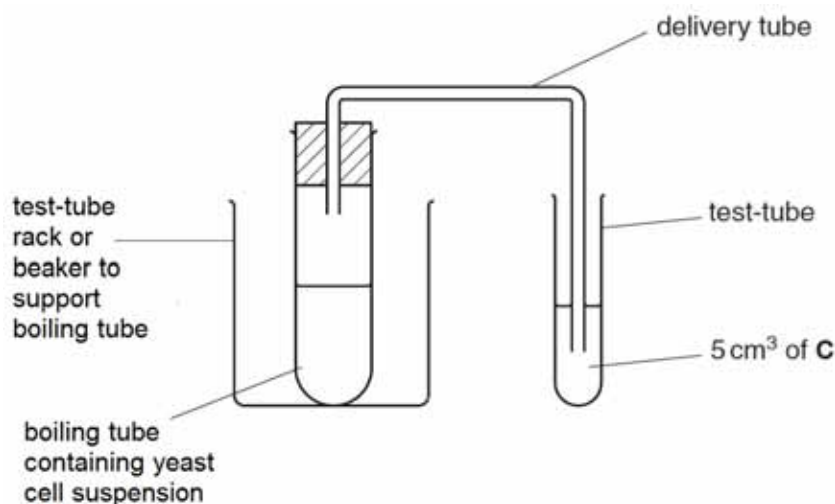
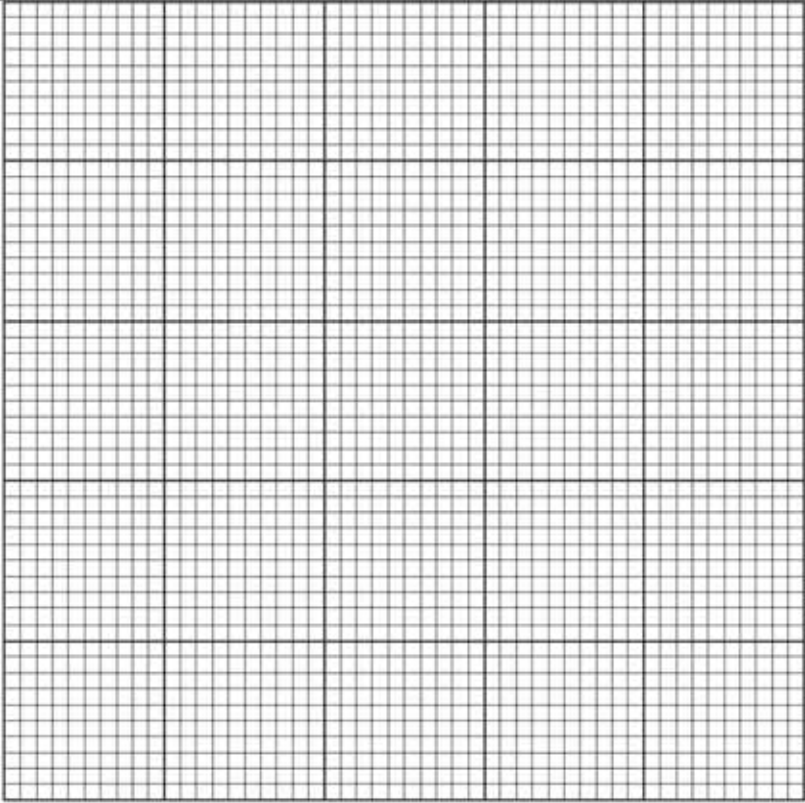


Fig.1.2

	(b)	State which sugar is expected to lead to a higher rate of carbon dioxide release by yeast cells. Explain your answer.	[2]								
	ACE Glucose 1. Readily used as a respiratory substrate by yeast;; 2. Unlike sucrose which must first be hydrolysed;; Sucrose 3. Hydrolysis of one molecule of sucrose produces two molecules of sugars;; 4. Higher concentration of respiratory substrates for respiration;;										
	Before proceeding with steps 6-12, ensure that 15 minutes have elapsed since stage 1.										
	6	Prepare and maintain a water bath at 40°C.									
	7	Label 3 test-tubes T1 , T2 and T3 . Add 5 cm ³ of C into each of the three test-tubes.									
	8	Place tube B1 in the water bath and allow it to equilibrate for 2 minutes.									
	9	Put the bung containing the delivery tube into tube B1 , as shown in Fig. 1.2. Ensure that the seal is airtight.									
	10	Put the end of the delivery tube into solution C in T1 . Using a stopwatch, allow the carbon dioxide produced by the yeast to bubble into C for 2 minutes.									
	11	After removing the delivery tube, repeat step 5 to determine the volume of H needed to reach end-point and record your result in (c) . If C has already reached end-point, record '0'.									
	12	Repeat steps 8 to 11 for the tubes B2 and B3 .									
	(c)	Record your results for each tube in a suitable form in the space below.	[4]								
	<u>Table showing effect of different substrates on the volume of H needed to reach end-point</u>										
	<table><tr><td>Type of substrate</td><td>Volume of H added / cm³</td></tr><tr><td>Glucose</td><td></td></tr><tr><td>Sucrose</td><td></td></tr><tr><td>Distilled water</td><td></td></tr></table>			Type of substrate	Volume of H added / cm ³	Glucose		Sucrose		Distilled water	
Type of substrate	Volume of H added / cm ³										
Glucose											
Sucrose											
Distilled water											

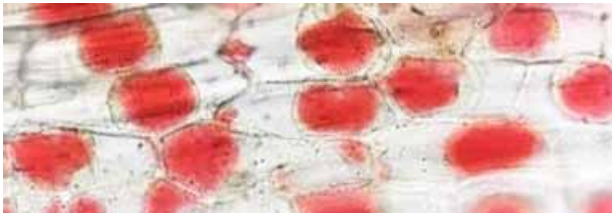
	PDO 1. Title (T);; "Table showing effect of different substrates on volume of H needed to reach end-point" 2. Heading with Units (H);; Independent variable: Type of substrate Dependent variable: Volume of H added / cm ³ 3. Precision (P);; Volume recorded to 1 d.p. MMO 4. Trend (Tr);; Volume of H added for B3 should be same as ≤ (a) and highest volume for B3		
	(d)	Discuss what these results suggest about the relationship predicted in part (b) .	[2]
		ACE 1. Comment on how actual results match the proposed relationship;; 2. Comment on how this affects the confidence/reliability in the proposed relationship / hypothesis ;;	
	The addition of 1 cm ³ of H into C is equivalent to 2.2 mg of carbon dioxide.		
	(e)	Calculate the exact amount of carbon dioxide produced by the yeast suspension from boiling tubes B1 and B2 during the experiment. Show your working clearly.	[2]
		ACE 1. Correct working;; Subtract reading in (c) from answer in (a) 2. Correct working; Multiply by 2.2mg 3. Correct calculation;	
	(f)	State the purpose of including boiling tube B3 in the investigation.	[1]
		ACE 1. To show that the evolution of carbon dioxide is due to the addition of glucose and sucrose / sugar;;	

	(g)	Identify one significant source of error and suggest a modification to the method that would overcome this.	[2]												
		ACE Error 1. Visual determination of colour change at end-point is subjective;; Modification 2. Use a colourimeter to determine precisely the colour intensity at end-point;;													
	In a separate study, a student used a modified version of the apparatus shown in Fig.1.2 to investigate the effect of temperature on the release of carbon dioxide from a yeast cell suspension. The results of the study are presented in Table 1.1.														
	Table 1.1														
	<table><tr><td>Temperature / °C</td><td>Volume of carbon dioxide evolved / cm³</td></tr><tr><td>25.0</td><td>93</td></tr><tr><td>30.0</td><td>126</td></tr><tr><td>35.0</td><td>160</td></tr><tr><td>40.0</td><td>180</td></tr><tr><td>45.0</td><td>156</td></tr></table>			Temperature / °C	Volume of carbon dioxide evolved / cm ³	25.0	93	30.0	126	35.0	160	40.0	180	45.0	156
Temperature / °C	Volume of carbon dioxide evolved / cm ³														
25.0	93														
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40.0	180														
45.0	156														
	(h)	Use the grid provided to plot the data given in Table 1.1.	[4]												

			
	PDO - Title (T); "Graph showing effect of temperature on volume of carbon dioxide evolved" - Heading with units (H&U);; x-axis: Temperature / °C y-axis: Volume of carbon dioxide evolved / cm³ - Precision (P); x-axis: 1 d.p. y-axis: whole no. - Scale (S); Graph takes up at least 50% of grid Divisions are equidistant - Plotted points (PP);; All data plotted accurately to within half a small square on the grid If the scale is awkward, the points do not need checking - Best-fit Line (L); Smooth curve of best fit with no extrapolation		
	(i)	Explain the relationship between the variables in yeast respiration.	[4]
		ACE - Respiration carried out by temperature-sensitive enzymes; As temperature increases, - enzymes and substrates gain more kinetic energy;	

		3. increase in frequency of effective collisions between enzyme and substrate; 4. increase in number of ES complexes formed per unit time; 5. more CO₂ given off per unit time; As temperature increase beyond optimum, 6. enzyme molecule vibrate so vigorously that the <u>hydrogen bonds</u> and <u>hydrophobic interactions</u> (between the R groups of amino acid residues) begin to break; 7. shape of active site not complementary to that of substrates / some enzymes denature; 8. less CO₂ given off per unit time;	
		[Total: 22]	

Question 2			
During this question you will require access to a microscope. You are required to investigate the effects of potassium nitrate and lead nitrate solutions on cells of the plant material with which you have been supplied. Peel off one or two strips of epidermis from the most deeply pigmented areas of the plant tissue. Remove as little of the underlying tissue as possible. Cut the epidermis so that you have two squares of tissue, each about 5 mm x 5 mm. Place these in a dish of distilled water. Mount one piece of tissue on a microscope slide in distilled water under a cover slip. Mount the other pieces in 1 mol dm⁻³ potassium nitrate solution. Label your slides appropriately. Examine the tissue mounted in distilled water, using your microscope. Find an area of the tissue where pigmented cells occur, preferably as a single layer.			
	(a)	Describe the distribution of the coloured contents within the cells.	[1]
		uniform distribution of coloured content within the cell;; 	
Observe the piece of tissue mounted in 1 mol dm ⁻³ potassium nitrate solution, using your microscope			
(b)	(i)	After about one minute, make a large drawing to show the detailed structure of one epidermal cell which is typical of the most deeply coloured cells which you can see. On your drawing, label the positions of: • a cell wall • a cell surface membrane.	[4]

			1. (T) Title (high power detailed drawing of one epidermal cell, w.m., X400); 2. (P) Thin walls; 3. (S) Shape (length longer than width) ; 4. (D) Cell membrane detached from cell wall; 5. Labels (cell wall, cell surface membrane) – 1 mark each 	
		(ii)	Account fully for the change in appearance of the cells when placed in 1 mol dm^{-3} potassium nitrate solution.	[3]
			1. 1 mol dm^{-3} sodium nitrate has a more negative water potential than the cell content; 2. water leaves cells; 3. by osmosis; 4. from a region of less negative water potential to a region of more negative water potential; 5. loss of water caused the cell surface membrane to pull away from the cell wall; 6. retention of pigments in the vacuole because of its high molar mass; 7. this phenomenon is known as plasmolysis; 8. the gap is filled with potassium nitrate solution;	
	Heavy metals such as lead and copper are toxic to plants.			
	(c)	Predict the appearance of the epidermal cells if the epidermis is mounted in 1 mol dm^{-3} lead nitrate solution. Explain your predictions.		[3]
		1. pigments leaked out of the cells / content dispersed;; 2. (heavy metal - lead) disrupts structure of membrane protein;; 3. loss of membrane selective permeability;;		
	Fig. 2.1 shows the view of a mammalian white blood cell, using an eyepiece graticule and the high-power objective lens of a microscope.			

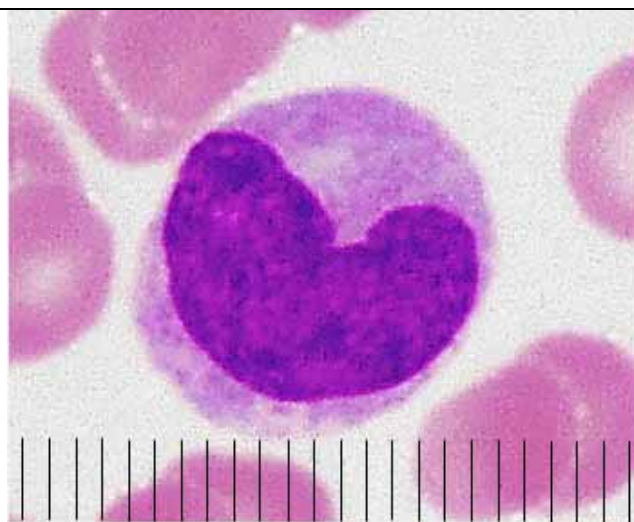


Fig. 2.1

	(d)	Use a table to record three observable differences between the blood cell in Fig. 2.1 and the cell you saw in (a) .	[3]
		1. Absence of cell wall;; 2. U shaped nucleus;; 3. Nucleus took up >50% of cell size;; 4. Nucleus in the centre of cell;; 5. Round shaped cell;;	
	(e)	(i)	
		The student calibrated the eyepiece graticule against a stage micrometer with the following results: Number of eyepiece graticule divisions across 5 stage micrometer division = 25 One stage micrometer division = 0.01 mm	
		Use this information to calculate the actual length of the cell. Show the steps and units in your calculation.	[3]
		1 division on eyepiece graticule = $(0.01 \times 5) / 25 = 0.002 \text{ mm} = 2 \mu\text{m};;$ Actual length of cell = 14;; x 2 = 28 $\mu\text{m};;$	
		(ii)	
		Calculate the magnification of the blood cell in Fig. 2.1. Show your working.	[2]
		Magnification = diagram size / actual size;; Magnification = 4.8 cm / 18 μm = X1714.3 (1 d.p.);;	
			[Total: 19]

Question 3

Industrial wastewater contains high concentrations of fats, like oil and grease, which may pollute fresh water and influence aquatic environments. The use of lipase enzymes to remove oil and grease from the wastewater is an effective and environmentally-friendly treatment method. Chemicals are added to the wastewater to emulsify the fats and break them up into smaller fat droplets, increasing the surface area for lipase to break them down into fatty acids and glycerol at a higher rate.

In wastewater treatment, enzymes are used on a large scale. However, enzymes are costly and most are only commercially available in liquid or dehydrated forms. As such, once added to the wastewater treatment mixture, they cannot be recovered to be re-used, thus driving up the cost of treatment.

Researchers postulated that the immobilisation of lipase in alginate beads would allow the enzymes to be re-used, leading to cost savings. The enzymes can be added to sodium alginate and immobilised in the beads as shown in Fig. 3.1.

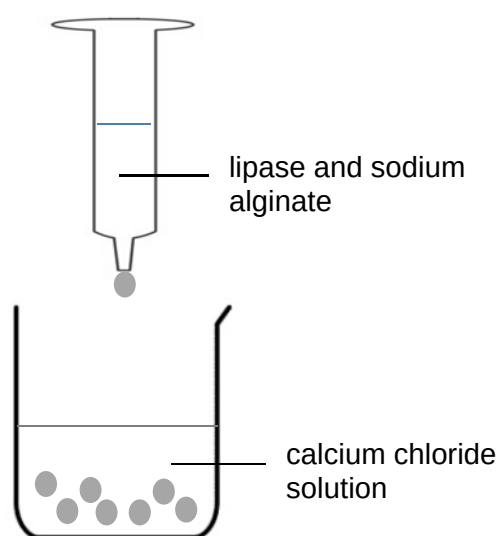


Fig. 3.1

Full-fat milk can be used as a substrate for the immobilised lipase enzymes. Bile salts act like detergents and can be used to emulsify the fats without affecting the pH. Sodium carbonate is a base that can be added to standardise the milk to an alkaline pH, prior to treatment with lipase.

The monitoring of the pH of the reaction mixture from a predetermined start to end-point pH can be used to measure the activity of the immobilised lipase.

Using this information and your own knowledge, design an experiment to find the concentration of lipase needed to form immobilised lipase beads with the same activity as free enzymes (non-immobilised).

You must use:

- 10% lipase solution
- industrial 3% lipase
- sodium alginate suspension
- calcium chloride solution
- full-fat milk
- sodium carbonate solution
- 5% bile salts solution

- a pH meter with a digital display to 2 decimal places
- stopwatch

You may select from the following apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rod, etc.
- syringes

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 14]

Relate independent variable to dependent variable

Reaction: Lipase is a biological catalyst which catalyses the hydrolysis of lipids into glycerol and fatty acids.

How to measure dependent variable

Fatty acid produced neutralises the alkali in milk, leading to a decrease in pH of the reaction mixture.

Time taken for pH to decrease from 10.00 to 7.00 can be used to measure the rate of reaction.

Describe and explain of expected trend

With the increase in concentration of lipase, the frequency of successful collision between substrate (lipids) and enzyme (lipase) increases. As more enzyme-substrate complexes are formed per unit time, more fatty acid formed per unit time. Hence time taken for the pH to decrease to 7.00 decreases.

Independent variable

Concentration of lipase

Range: 10%, 8%, 6%, 4%, 2%

Dependent variable

Time taken for the pH to decrease from 10.00 to 7.00

Controlled variables

Controlled Variable	Appropriate Quantity
Temperature of incubation	40°C
Volume of milk used	5 cm ³
Volume of lipase used	2 cm ³
Volume of bile salts	1 cm ³
Volume of sodium carbonate	5 cm ³

Independent variable:

States that independent variable is concentration of lipase and uses at least five different uniformly-spaced concentrations;;

Dependent variable:

Time taken for pH to reach stated end-point;;

Controlled variables:

Describe two identified variables;;

And describe how they are controlled;; ;

Experimental Procedures

Dilution of lipase:

Table showing dilution 10% lipase solution

Final concentration of lipase / %	Final volume of diluted lipase / cm ³	Volume of stock added / cm ³	Volume of distilled water added / cm ³
10	5.0	10.0	0.0
8	5.0	8.0	2.0
6	5.0	6.0	4.0
4	5.0	4.0	6.0
2	5.0	2.0	8.0

Preparation of immobilised lipase:

1. mix 2 cm³ of lipase with 2 cm³ of sodium alginate
2. Pour the mixture into a syringe
3. Drip mixture into calcium chloride solution
4. Wash the beads with distilled water
5. Repeat for each concentration of lipase

Set up for immobilised lipase:

1. Set up at water bath at 40°C using a 500 cm³ beaker
2. add 5 cm³ of milk into a test-tube
3. Add 1 cm³ of bile salts solution to emulsify the fats
4. Using the pH meter, add sufficient (e.g. 5 cm³) sodium carbonate solution so that the pH of the solution in the test-tube reaches pH 10.
5. Place all the lipase beads into another test-tube
6. place both tubes in the water bath 40°C for 2 minutes to equilibrate to temperature
7. put in the pH probe and ensure the reading is 10.00
8. transfer the 5 cm³ milk into the test tube containing immobilised lipase and start timing
9. Record the time taken for the pH to reach 7.00
10. Repeat steps 2 – 9 for each concentration of lipase enzyme

Set up for free lipase:

Same procedure as above except that 2 cm³ of lipase is added into a test tube

Producing accurate and reliable results

- For each concentration of lipase enzyme, repeat the entire procedure to obtain three replicates
- Repeat the entire experiment two more times

Definition of control Negative control

A negative control is subjected to the same factors as that for the experiment, except that lipase is replaced by an equivolume of distilled water

It is expected that the pH remains at 10.00 regardless of the incubation time.

This proves that it is indeed lipase that hydrolyses the fats causing pH to decrease.

Data manipulation

Method:

Plans suitable method to vary lipase concentration;;

Method:

Plan suitable method to immobilise lipase in beads

Method:

Devise a method that equilibrate the enzymes and substrate to temperature;;

Method:

Standardise the starting pH of mixture e.g. pH 10 by adding a fixed volume of sodium carbonate;;

Method:

Plans a suitable procedure that involves monitoring the change in pH towards a set end-point pH;;

Method:

Plan a method to carry out the same procedure but with equal volume of free enzyme

Reliability:

Ref to repeating at least two more times with different lipase concentration;;

Control:

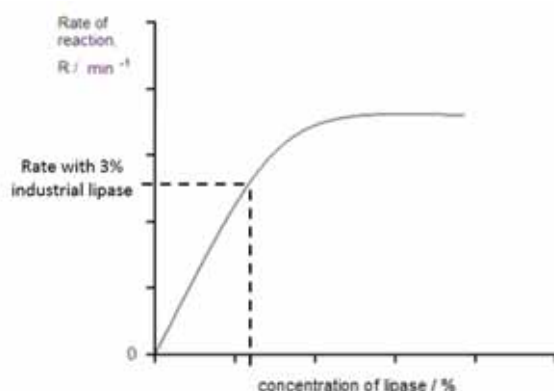
Ref to control without substrate;;

Processing of data – mean / average
rate of reaction = $1/t$

Table showing the effect of concentration of lipase on rate of reaction

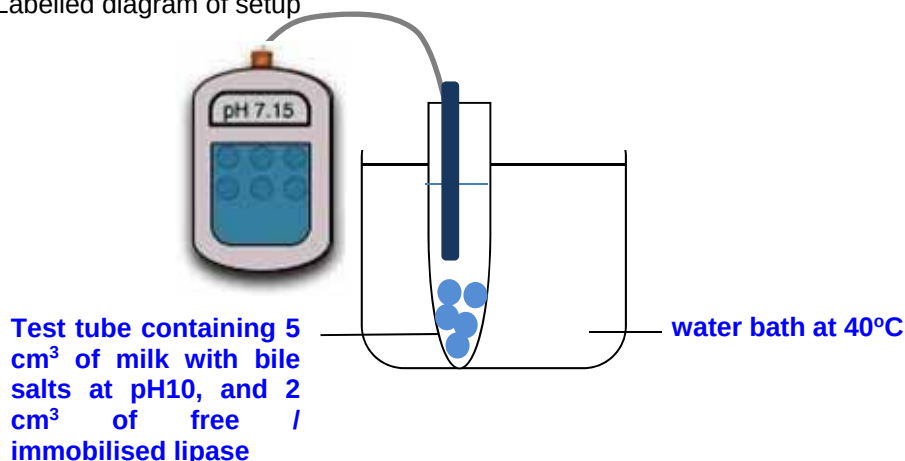
Lipase concentration, C/ %	Time taken for pH to change from 10.00 to 7.00, t/ min				Rate of reaction, R/ min ⁻¹
	t_1	t_2	t_3	$\bar{t}$	
10					
8					
6					
4					
2					

Graph showing the effect of concentration of lipase on rate of reaction



To find out the concentration of lipase to make immobilised enzymes, plot a standard curve with the data obtained. The concentration of lipase to use is then obtained by using the graph to determine the lipase concentration that gives the same rate of reaction as 3% industrial lipase (free enzymes).

Labelled diagram of setup



Risk assessment and precautions taken

- Wear gloves and goggles to prevent contact with lipase which is an irritant.
- Use a piece of rag to hold the beaker containing hot water (when

Recording:

Shows how results are to be presented in the form of a table with IV (concentration of lipase) and dependent variable (time taken for pH to change from 10.00 to 7.00) in appropriate column/rows;

Accuracy:

States how lipase concentration can be determined by comparing to reaction by free enzymes

Risk/safety:

Refers to hazard and precaution;;

preparing water bath) to prevent burns / scalds Correct use of technical and scientific terms e.g. catalyse, enzyme-substrate complex, active site, complementary, acidic	
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